

A Three-Level Automated QA/QC Framework for Geotechnical Instrumentation Data in TSFs

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ABSTRACT

Geotechnical monitoring of tailings storage facilities (TSFs) generates continuous data records from extensive instrument networks. However, the reliability of that data is rarely subjected to systematic validation. The predominant industry practice remains reactive: inconsistencies are frequently identified manually, with significant delays, and quality controls are applied without consistent criteria across instruments and projects.

This paper proposes a three-level automated data quality assurance and quality control (QA/QC) framework for geotechnical instrumentation data in TSFs, designed to transform raw monitoring records into reliable, traceable, and decision-ready information. The framework operates across three hierarchical levels: (N1) integrity validation, which verifies that data arrived complete, in the correct format and without gaps relative to the expected acquisition frequency; (N2) physical validation, which checks that readings fall within reliable ranges for the instrument and site; and (N3) contextual validation, which assesses the consistency of each reading against the instrument's historical behavior and neighboring instruments. It is argued that robust data validation is not an optional improvement but a fundamental requirement for transparent and auditable monitoring under the Global Industry Standard on Tailings Management (GISTM).

INTRODUCTION

Monitoring of tailings storage facilities (TSFs) has evolved considerably over recent decades: sensors are more precise, data acquisition systems more reliable (Dunnicliff, 1993), and remote access to near-real-time information is increasingly common. The geotechnical integrity of these facilities depends critically on the quality of the monitoring data used to support operational decisions (Fell et al., 2015). However, these advances have not been accompanied by equivalent development in the processes of data validation and quality assurance and control (QA/QC).

In most monitoring programs, validation is performed manually by engineers reviewing monitoring platforms, dashboards, or spreadsheets, without documented criteria for data acceptance or rejection. This approach is not scalable as instrument networks grow, and is neither traceable nor auditable, since no systematic record exists of what was verified, when, and against what criteria. This gap is particularly significant in the context of the GISTM (ICMM, PRI & UNEP, 2020), whose Principle 7 requires designing and implementing monitoring systems adequate to verify design assumptions (Requirement 7.2), with specific and measurable performance criteria and parameters (Requirement

7.3), and whose Principle 15 requires public disclosure of monitoring program findings. These requirements presuppose that data quality is demonstrable; yet the standard does not define specific validation methods.

This paper proposes an automated three-level validation framework, implementable within existing monitoring platforms and independent of instrument type or communication protocol. The paper first presents the data quality problem in geotechnical monitoring and a classification of the most frequent failure types, followed by the framework's development, its implications for governance and regulatory compliance, and considerations for implementation.

METHODOLOGY

The data quality problem in geotechnical monitoring

Instrumentation technical specifications focus on instrument procurement and installation, while geotechnical monitoring and control plans (including Trigger Action Response Plans (TARPs)) define reading frequencies, operational thresholds, and response protocols for events that could compromise facility stability. However, criteria for determining whether a data point is valid, uncertain, or rejected prior to its use in decision-making are rarely defined. When such criteria exist, they are typically formulated in qualitative terms, such as: "readings must be within expected ranges", without defining those ranges or specifying actions to take when they are exceeded, creating a structural vulnerability: the system continuously collects data, but cannot distinguish valid readings from instrument errors or physically inconsistent values.

Classification of data quality failures

Based on operational experience with geotechnical instrumentation in TSFs, data quality failures are classified into five categories:

- **Transmission failures:** missing data, data with errors, or data with incorrect date/time stamps. In automated instrumentation, these are detected without analyzing the recorded value. In manual instrumentation, the system must distinguish between expected gaps by design (intervals between field campaigns) and unplanned gaps due to failure.
- **Slow sensor drift:** gradual and sustained deviation without an external event to justify it, where each individual reading appears reasonable and is only detectable through trend analysis over time.
- **Abrupt jumps or step changes:** sudden large-magnitude change between consecutive readings, at a rate physically impossible for the monitored variable. Detected by comparing the rate of change against thresholds defined per instrument and site.
- **Out-of-range or physically inconsistent values:** readings outside the instrument's measurement range, inconsistent with the measured variable at that location, or within the nominal range but below the sensor's minimum resolution (noise).

- **Cross-instrument inconsistency:** individually reasonable readings that are inconsistent with nearby instruments, suggesting a localized deviation in the sensor rather than in the monitored medium.

Three-level validation framework

The three levels are designed to be sequential (a reading that fails N1 does not require evaluation at N2), but each level can be implemented independently according to the project's technical capabilities and available data history.

Table 1 Summary of the three-level validation framework

Level	Validation type	Detected failures	Automation potential
N1	Integrity	Transmission failures	Fully automatable
N2	Physical	Abrupt jumps; out-of-range or physically inconsistent values	Rule-based, configurable per instrument
N3	Contextual	Slow sensor drift; cross-instrument inconsistency	Statistical or anomaly-detection models

Level 1 (N1) — Integrity validation

Verifies data receipt, compliance with the expected format, and temporal consistency with the configured acquisition frequency. Checks include: reading completeness, timestamp consistency, numeric format validity, and transmission metadata (signal level, battery status). Since it operates exclusively on data acquisition system configuration parameters, no geotechnical knowledge is required, and it constitutes the minimum validation standard for a TSF under the GISTM.

Level 2 (N2) — Physical validation

Assesses the physical consistency of each reading based on three instrument-configurable parameters: the sensor's measurement range, the expected limits of the variable at a specific location, and the maximum admissible rate of change. Site limits incorporate the expected range of the monitored variable given the facility's geotechnical conditions (e.g., phreatic level consistent with embankment geometry) and design criteria; once defined, these parameters operate automatically and should be reviewed annually or following significant site changes.

Level 3 (N3) — Contextual validation

Evaluates the consistency of each reading against the instrument's history and neighboring instruments, enabling detection of deviations that are not identifiable during physical validation because each individual reading would meet physical consistency criteria. Checks include: trend analysis against a historical baseline, where sustained deviations generate a record quality deterioration indicator, and spatial consistency among neighboring instruments, where a deviation without correlation between neighboring sensors suggests a localized failure. For most projects, rule-based statistical approaches (e.g., linear regression over time windows) are sufficient without resorting to machine learning models. This level requires a minimum of six to twelve months of validated data history from the preceding levels.

Data quality states as a result of the validation process

As a result of the validation process, each reading is classified into one of three states: validated, when it meets the criteria of all three levels; flagged or uncertain, when it presents deviations requiring review by the geotechnical engineer in charge of monitoring; or rejected, when it fails criteria that render it invalid for use in decisions.

Proof-of-concept validation

To validate the completeness of the proposed framework, a synthetic time series was constructed for a piezometer in a TSF (baseline ~1.5 mwc, 140 hourly readings) with nine failure scenarios covering each of the five failure categories and all three validation levels, demonstrating that no failure category is left unaddressed by the three levels combined. Figure 1 presents the resulting time series and the nine embedded scenarios:

- ① **Transmission failure (N1):** five-hour data gap due to loss of communication; an instrument offline alert is generated.
- ② **Abrupt jump or step change (N2):** 3.5 mwc increase in one hour, exceeding the maximum admissible rate of change.
- ③ **Slow sensor drift (N3):** sustained increase of 0.06 mwc/h over 20 hours, detectable only through historical trend analysis.
- ④ **Negative value due to incorrect configuration (N2):** -2.8 mwc reading due to an incorrect zero offset in the datalogger.
- ⑤ **Sensor saturation (N2):** sustained reading of 9.9 mwc, corresponding to the sensor's maximum measurement limit, detected by comparison against the instrument's maximum physical range.
- ⑥ **Spike or erroneous point reading (N2):** isolated reading of 8.2 mwc (~1.7 mwc), attributable to electrical interference or instrumental error; detectable by N2 (site range exceedance) and confirmable by N3 (instrument's recent history).
- ⑦ **Negative value due to real suction (N2+N3):** moderate and transient negative value requiring spatial coherence verification to distinguish between real suction and

instrumental error; N2 alerts for exceeding the normal range and N3 determines validity by verifying coherence with neighboring instruments.

- ⑧ **Progressive drift toward negative (N3+N2):** gradual descent from positive values to below zero; N3 detects the trend before N2 invalidates the reading upon reaching the physical limit.
- ⑨ **Frozen reading (N3):** constant value of 1.58 mwc for nine consecutive hours; N3 detects zero variance, indicating the datalogger is repeating the last stored reading.

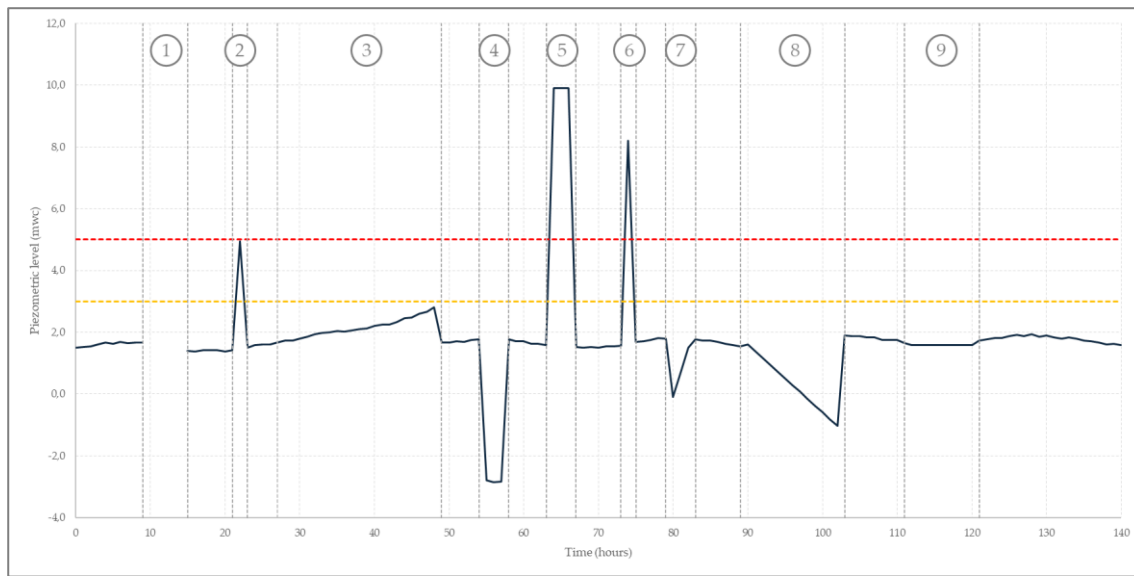


Figure 1 Synthetic time series for a piezometer in a TSF with nine failure scenarios covering the five failure categories and three validation levels; reference lines: caution threshold (yellow dashed) and high threshold (red dashed)

RESULTS AND DISCUSSION

Implications for governance and regulatory compliance

Although the GISTM (ICMM, PRI & UNEP, 2020) does not define specific validation methods, its transparency requirements (independent reviews, public disclosure, and audit records) imply that data quality must be demonstrable and systematic. The proposed framework responds to this requirement by assigning each reading a quality status, the validation level at which it was classified, and the criteria applied, thereby creating a traceability chain from the raw sensor data to the decision or action. This approach extends the data quality assurance work of Magalhães et al. (2023), who address QA/QC during monitoring system commissioning, into the operational phase of continuous data acquisition. It also complements the anomaly identification methods of Dong et al. (2022), whose statistical and neural network approach to seepage monitoring data in TSFs aligns with the contextual validation objectives of Level 3 (N3) of this framework.

Implementation considerations

The framework allows for gradual adoption compatible with existing data acquisition systems:

- **Level 1 (N1):** can be implemented immediately, as it operates exclusively on the existing system configuration; it is the recommended starting point.
- **Level 2 (N2):** requires knowledge of instrument technical capabilities and threshold definition in collaboration with the geotechnical engineer in charge of facility monitoring, after which it operates automatically.
- **Level 3 (N3):** requires prior consolidation of the preceding levels and a minimum of six to twelve months of validated data as a baseline.

Across all three levels, the system classifies and flags, but the final decision on flagged or rejected readings rests with the geotechnical engineer in charge of monitoring.

CONCLUSION

This paper presents a three-level automated QA/QC framework for geotechnical instrumentation data in tailings storage facilities. The main contributions are: a classification of five failure categories spanning from transmission errors to cross-instrument inconsistencies; a hierarchical validation framework implementable progressively and compatible with existing data acquisition systems and monitoring platforms; and a data classification scheme that assigns each reading a quality status, the validation level, and the criteria applied, forming an auditable record consistent with GISTM transparency requirements.

The framework allows for gradual adoption according to each project's capabilities, and across all its levels considers one central principle: the system classifies and flags, but decisions on uncertain data or data that does not meet expected quality rest with the geotechnical engineer in charge of monitoring. Automated data validation is not an optional improvement but a necessary component for geotechnical monitoring to effectively fulfill its role in TSF safety management. Future work should include validation against real operational datasets to quantify detection rates, false positive ratios, and computational performance across different instrument types and site conditions, which would further consolidate the framework's applicability in operational monitoring programs.

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